

Debayan Biswas

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Backend engineer specializing in REST API design, distributed systems, and PostgreSQL-driven services.
Production experience in Java microservices, async data pipelines, Redis caching, and OAuth 2.0. IIT Kharagpur.

EDUCATION

Indian Institute of Technology, Kharagpur

Bachelor of Science, Physics

Kharagpur, West Bengal

Apr. 2025

WORK EXPERIENCE

Software Engineer

Cimba.ai

May 2025 – Present

Kolkata, West Bengal

- Defined REST API dispatcher and analytics endpoints for KPI and trend metrics (latency, tokens, failures, users, tasks) with server-side pagination and presigned CSV export; released in production with React/ECharts dashboard.
- Integrated OAuth 2.0 with 3 token flows (entity-scoped, sharing, refresh), Redis-powered caching for API key validation and chat sessions, and an async data ingestion pipeline from Google Sheets into DuckDB.
- Engineered real-time WebSocket event system for Deep Research across 3 event types (memory, notifications, completion); propagated RequestId across microservices for end-to-end distributed tracing.
- Delivered Enrich REST API supporting JSON, CSV, and XLSX ingestion with async migration and relational-table enrichment; launched server-side agent-builder table APIs, retiring a legacy email bypass.
- Rearchitected search infrastructure: replaced Pinecone with Azure AI Search, developed Java-based agentic runtime with async knowledge-base indexing; deployed migration pipeline with zero-downtime cutover.
- Refactored LLM regression test suite from 700-line Python monolith to modular Java, cutting configuration duplication 71%; owned CI/CD pipeline with three-channel Slack alerting across 5 LLM providers.

Software Engineer Intern

Ittiam Systems

Jun. 2024 – Jul. 2024

Bangalore, Karnataka

- Fixed A/V sync in a production video pipeline using FFmpeg split-and-stitch over chunked media.
- Architected and operated a 3-tier distributed video encoding pipeline: NFS-mounted shared storage, KEDA-driven autoscaling, Redis/BullMQ workers scaled dynamically on queue depth.
- Provisioned a 3-component Minikube Kubernetes dev stack and Docker images for repeatable integration tests.

Software Engineer Intern

OneLot

Dec. 2023 – Apr. 2024

Manila, Philippines

- Integrated Slack for cross-functional team communication (15% collaboration lift) and partnered on SEO initiatives driving 25% increase in organic traffic.

PROJECTS

IntentFlow | Go, PostgreSQL, REST API, OpenAPI, JWT, RBAC, Docker, CI/CD

Feb. 2026

- Implemented a Go REST API service with OpenAPI contract: intent/plan/confirm workflow on PostgreSQL with JWT/RBAC multi-tenant isolation; DAG executor with parallel steps and dependency resolution.
- Integrated multiple LLM backends (OpenAI, Gemini, Hugging Face) via unified registry; added MCP orchestration, Dockerized PostgreSQL, CI pipelines, and end-to-end test coverage (73 commits).

TECHNICAL SKILLS

Languages: Java, Python, Go, TypeScript, JavaScript, Scala, C/C++

Backend & APIs: REST APIs, microservices, distributed systems, system design, event-driven workers, WebSockets, OpenAPI

Databases: PostgreSQL, Redis, SQL, DuckDB, Azure AI Search

Cloud & Infra: AWS, Azure, Docker, Kubernetes, CI/CD, Git, Linux, Agile/Scrum, DataDog

Security & AI: OAuth 2.0, JWT, RBAC, MCP, LLM APIs